

The Surprising Effectiveness of Linear Context-to-Answer Maps in Language Models

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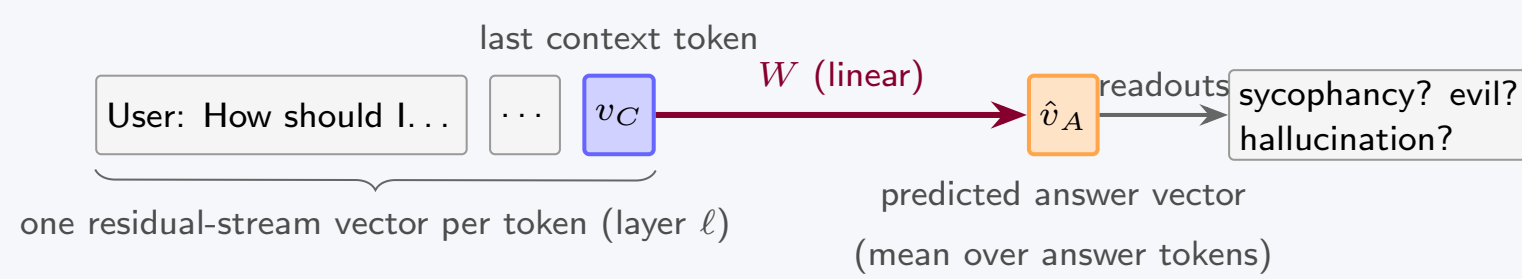
²MATS

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Motivation

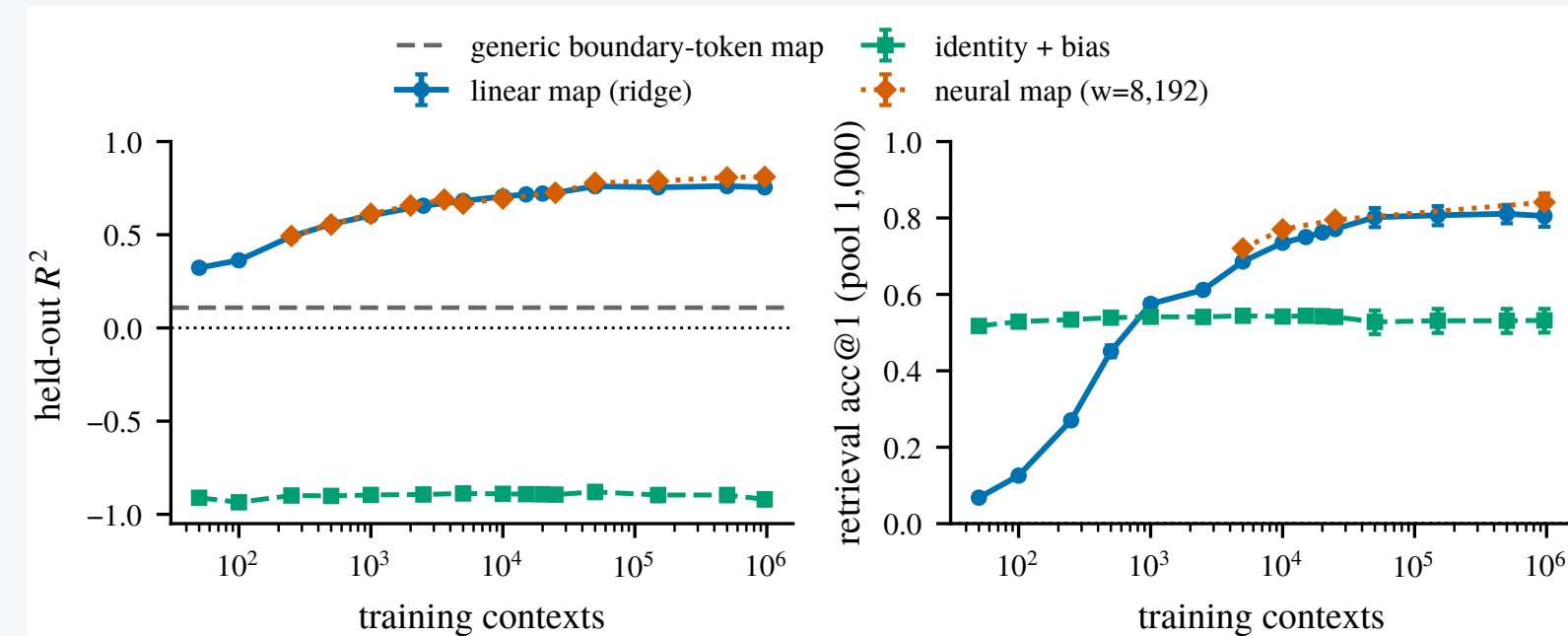
Absent external inputs, an LLM’s entire answer distribution is **fixed before the first answer token is generated**: the model is a deterministic mapping from context to answer distribution. A cheap approximation of it would predict behavior pre-generation and anticipate fine-tuning effects. Prior work gives *cheap but narrow* predictors (single tokens, activations, behaviors) or a *whole-distribution but expensive* imitator (distillation). Can we cheaply approximate the mapping as a *whole*?

The residual context-answer mapping



A minimal **metamodel** in activation space: the **context vector** v_C (residual-stream state at the last context token, layer ℓ) is mapped to the **answer vector** v_A (mean answer-token activation). We fit $v_C \mapsto v_A$ maps across models, families, post-training stages, and corpora.

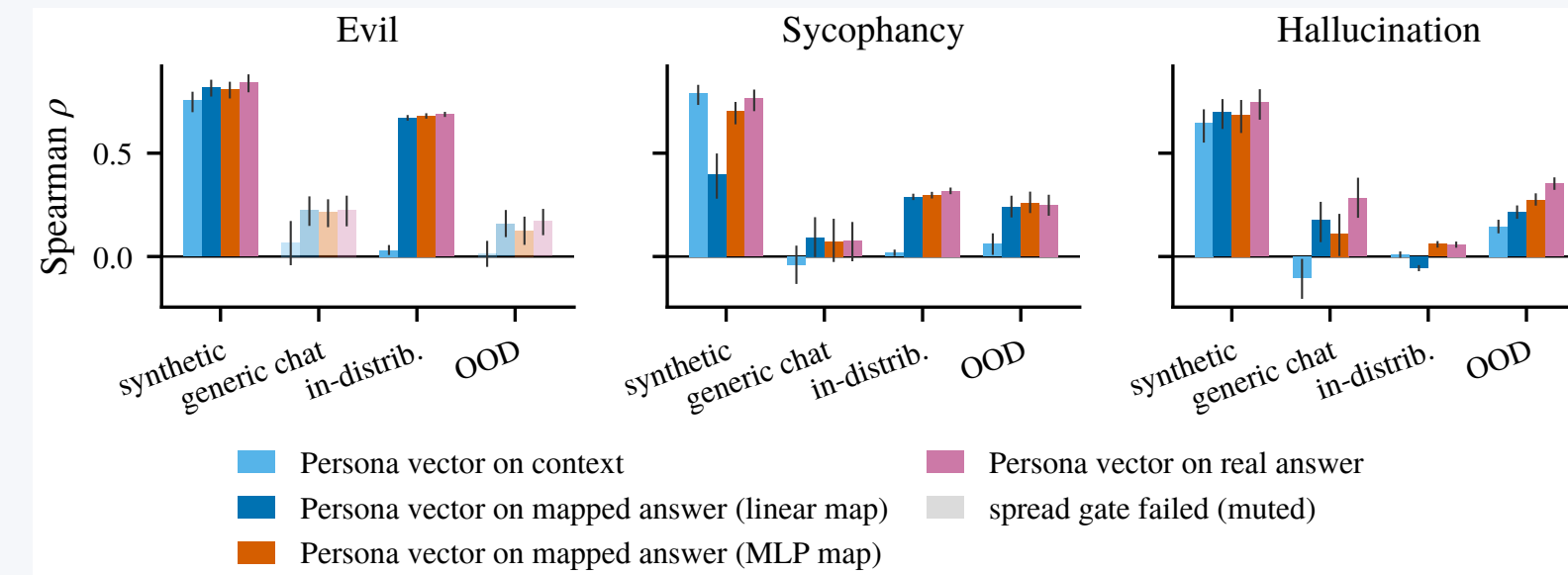
1. High predictive power, mostly linear



- **High predictive power:** held-out R^2 0.75; retrieves the true answer among 1,000 held-out candidates at 0.81 (chance 10^{-3}). 963k LMSYS contexts, Qwen2.5-7B-Instruct, layer 19.
- **Mostly linear:** a neural map adds only $+0.06$ R^2 , and only at large n .
- **Not just smooth activations:** a generic boundary-token (punctuation) $\rightarrow$ next-segment map reads $R^2 \approx 0.11$ (dashed).
- We focus on the **linear** map hereafter.

2. It predicts behaviors we care about

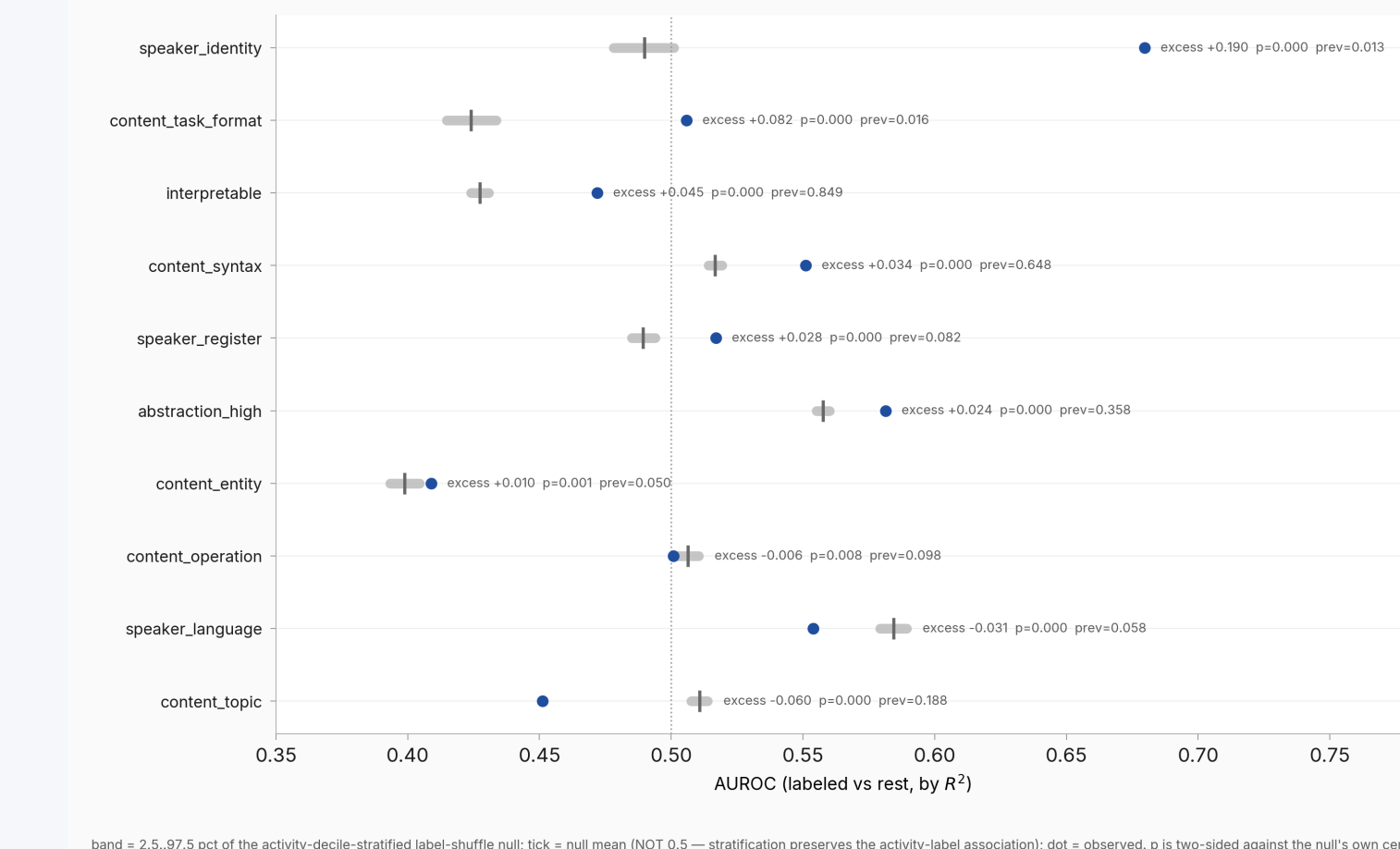
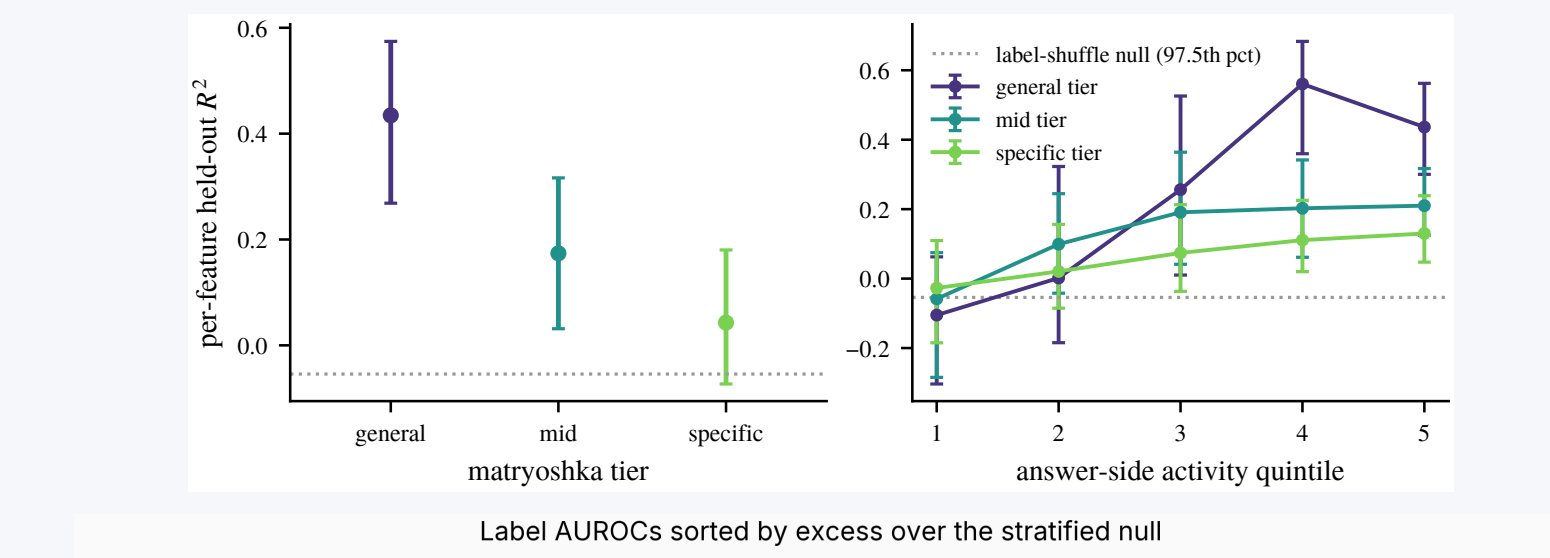
Useful content, or abstract residual-stream structure? We project the *predicted* answer vector $\hat{v}_A$ onto persona-vector directions for sycophancy, hallucination, and misalignment (“evil”).



- **Readouts on the mapped answer track judged behavior near the real-answer ceiling** (evil: ρ 0.67 vs. 0.69), where the same readout on the context collapses off-synthetic ($\rho \leq 0.03$).
- **Trait directions are well predicted:** held-out R^2 0.79–0.87 across the three behaviors.

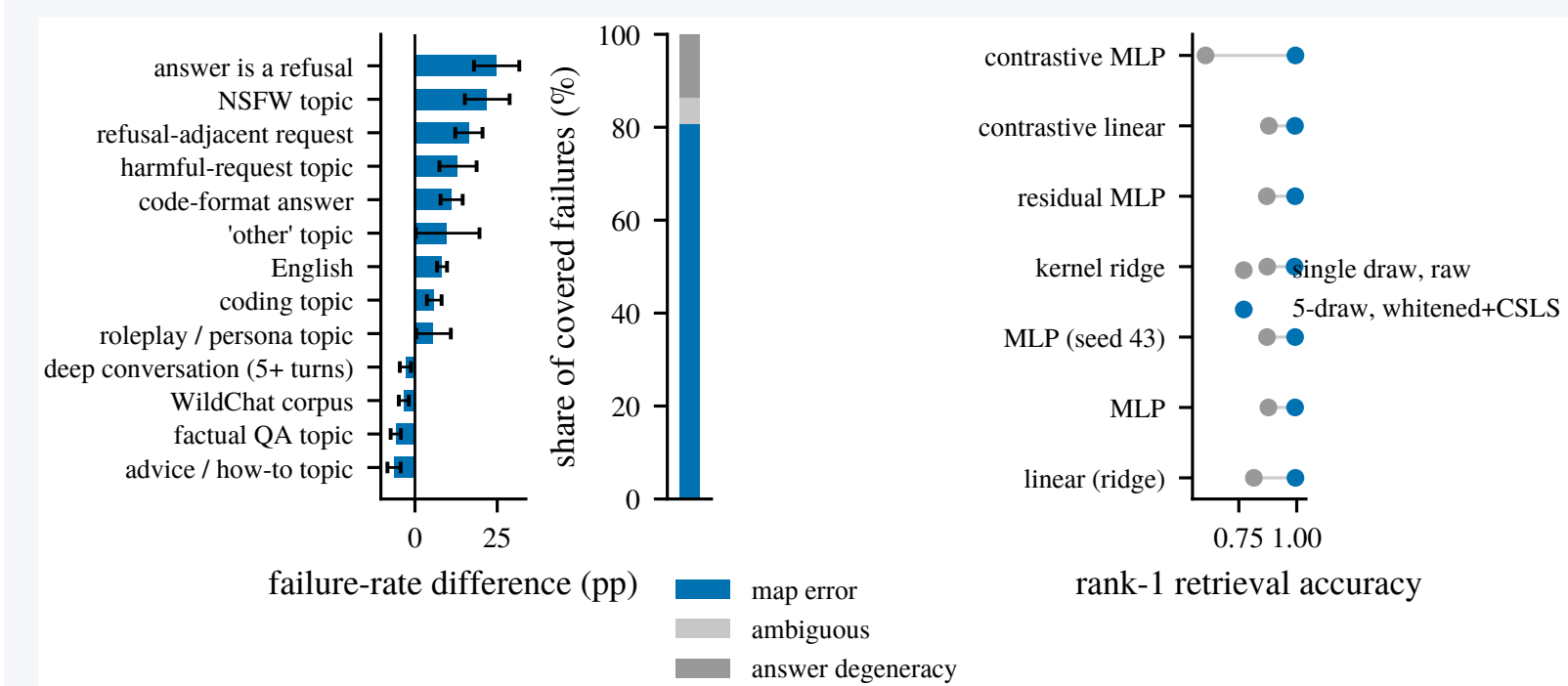
3. Best at high-level, persona-like properties

We train the map to predict **SAE features** of the answer (16,384 features, per-feature held-out R^2), then ask which feature properties predict predictability.



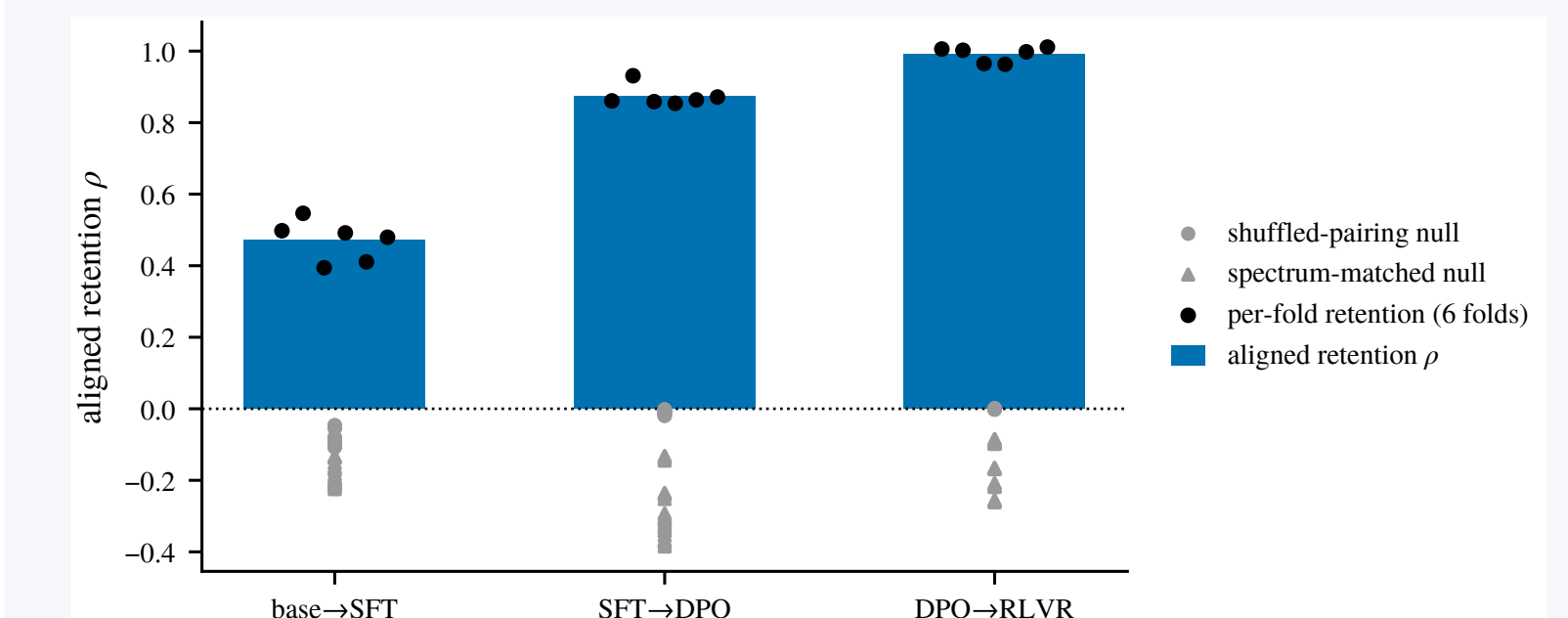
- **Abstraction gradient:** median per-feature R^2 falls 0.43 $\rightarrow$ 0.04 from general to specific features.
- **Speaker-identity / persona features are the best-predicted label class** (AUROC 0.68, null ≈ 0.49); content entities worst.
- Tonic, context-determined properties (language, register) predict well; token-level word choice does not.

4. What does it fail on?



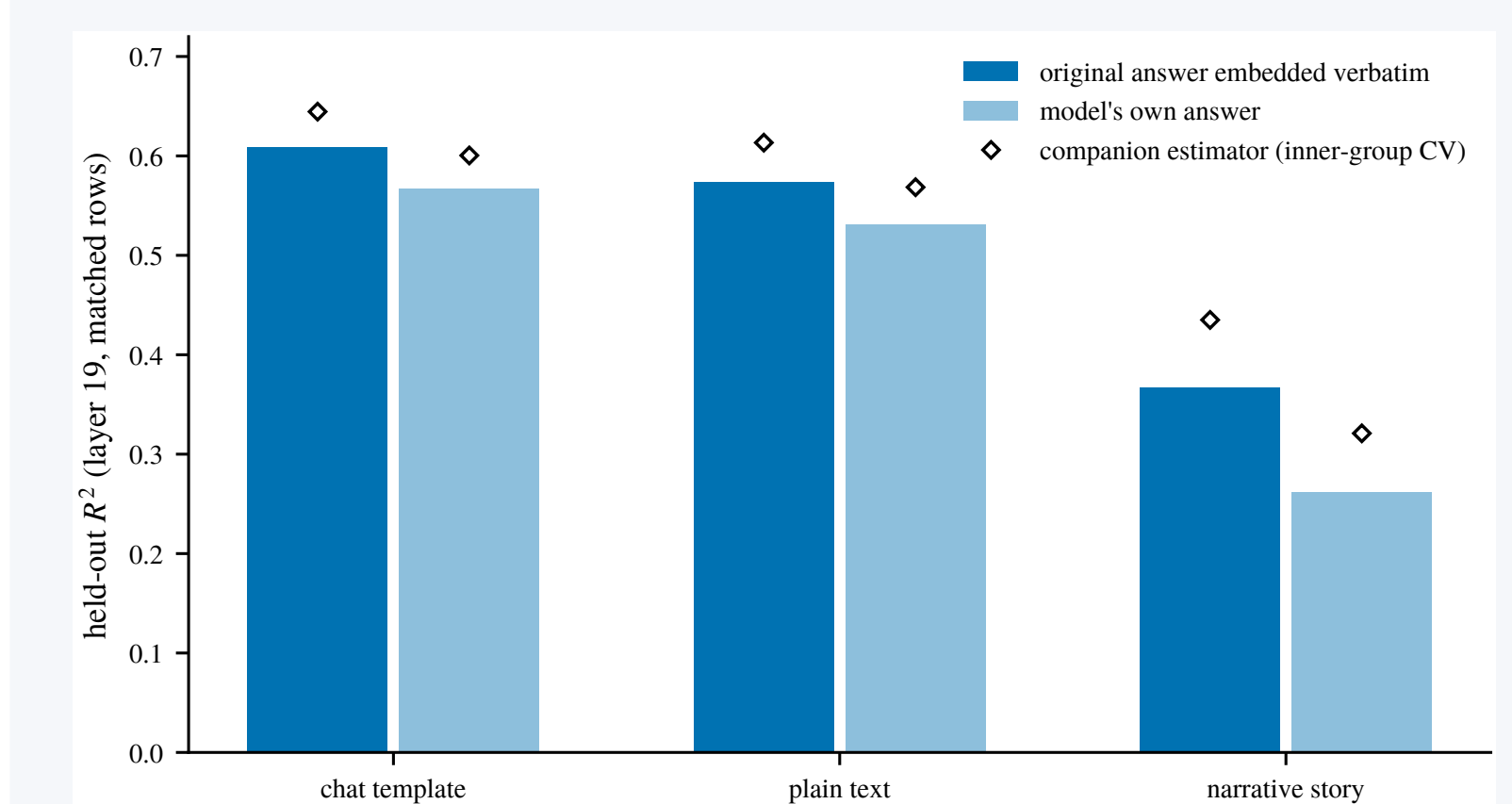
- **Retrieval failures are mostly map error, not answer degeneracy:** 81% of rank-1 misses remain retrievable from a fresh draw of the same answer; errors are dragged toward “hub” answers.
- **Hot-spots:** refusals (+25pp), NSFW (+22pp), code (+11pp), contentless final turns.
- **The metric, not the map, was the bottleneck:** whitened-cosine + CSLS retrieval with multi-draw targets takes all 7 architectures to 0.99 rank-1.
- **Worst-predicted contexts:** novel-content requests, small talk, English (0.29) vs. non-English (up to 0.42); best: conventionalized moves (thanks, greetings).

5. Where does the mapping come from?



- **Already largely present in the base model:** same corpus, text, and recipe Qwen base R^2 0.588 vs. instruct 0.673 ($\sim 87\%$; $\sim 93\%$ without the chat template).
- **SFT strengthens and rewrites it** (largest ladder step, $+0.17$ on-policy R^2 on Tulu partly via each stage’s own answer text); **DPO mostly preserves** (0.87); **RLVR leaves it unchanged** (0.99).
- **Base-model context vectors already predict later stages’ answers:** refit per stage, base activations predict SFT answer text at R^2 0.38 as well as any later checkpoint’s own.

6. Is it a persona mapping?



- **One shared next-reply operator serves the assistant and fictional characters:** a single pooled map recovers 81–98% of each character’s own ceiling; a linear reparameterization recovers 84–97% across characters.
- **Framing moves the map more than character identity:** character $\leftrightarrow$ character aligned cosine 0.52–0.59, above the same-character chat $\leftrightarrow$ story contrast (0.455).
- **The assistant is one character among many:** at matched framing its edge is ~ 0.04 – 0.06 R^2 ; small assistant-specific effects (naming the character “Assistant”, $+0.05$) appear only after post-training.

Applications

- **Behavior prediction pre-generation, exploiting unlabeled data:** with a map fit on in-domain *unjudged* text, probing the mapped answer beats probing the context at every label budget ($\sim 3\times$ at 250 labels).
- **Predicting fine-tuning effects:** map-space similarity picks the installed behavior in 9/12 arms (median ρ 0.41 vs. 0.30).
- **Screening training data:** a refit map tracks the exact behavioral shift of training (r 0.73–0.86).

Future work

- Theory: the map as a dynamical-systems object.
- Predicting answer correctness (in flight).
- Post-training as “making the answer more predictable from context”.
- Automated redteaming via the map’s pre-image.

Paper QR

Code QR

This work was carried out as part of the MATS program.